Q. Define Promoter?

A promoter is the DNA sequence that initially binds to the RNA polymerase with IF (Initiation Factors)

Q. what is called Closed Complex in initiation?

The first step is the initial binding of polymerase to a promoter to form what is called closed complex

What is escaped the promoter?

Ans Once an enzyme makes a transcript loner than 10 nucleotides, it is said to have escaped the promoter.

Q. what component or ingredients ternary complex contains?

Ans ternary complex containing enzyme, DNA and RNA this is the transition to the elongation phase.

Q. during Elongation what tasks enzymes performed in addition to catalysis of RNA?

1) it unwinds the DNA in front and re-anneals it behind.

2) it dissociates the growing RNA chain from the template as it moves along

3) it performs the proofreading functions

Q. What is called termination? in transcription process?

Ans. Once the polymerase has transcribed the length of the gene it must stop and release the RNA products. this step is called termination.

Q. Define RNA polymerase Holoenzyme

Ans. Sigma factor coverts core enzyme (Alpha2 BB w) into the form that initiates only at promoters this form of the enzyme is called the RNA polymerase holoenzyme.

Q. Define Isomerization z

Ans. In case of teh bacterial enzyme bearing sigma70 , this transistion is often called isomerization

Q. Define Transient Excursion model.

Ans The transient excursion model proposes transient cycles of forward and reverse translocation of RNA polymerase thus, polymerase is thought to leave the promoter and translocate a short way along the DNA template, synthesizing a short transcript before aborting transcription, releasing the transcript, and returning to its original location on promoter.

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Q. Define the Inchworming model.

Ans inchworming invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site. The module moves downstream, synthesizing a short transcript before aborting and retracting to the body of the enzyme still at the promoter

Q. Define The scrunching model?

Ans. Scrunching proposes that DNA downstream from the stationary, promoter-bound, polymerase is unwound and pulled into the enzyme the DNA thus accumulated within the enzyme is accommodated as single-stranded bulges.

Q. Define the RNA polymerase proofreading functions.

RNA polymerase performs two proofreading functions on growing transcript, the first of these is called **pyrophosphorolytic editiing** in this the enzyme uses its active site in a simple back-reaction, to catalyze the removal of an incorrectly inserted ribonucleotide, by reincorporation of PPi. the enzyme can then incorporate another ribonucleotide in its place in the growing RNA chain.

in the second proofreading mechanism called **hydrolytic editing,** the polymerase backtracks by one or more nucleotides and cleaves the RNA product, removing the error-containing sequence

Q what is the function of Nus Proteins?

Ans The Nus Protein join polymearase in the elongation phase and promotes the processes of elongation and termination

Q. What is called terminators?

Ans The sequences called terminators trigger the elongating polymerase to dissociate from the DNA and release the RNA chain it has made

Q. Explain 2 types of terminators in Bacteria?

1. Rho-Dependent

2. Rho-independent

The first, requires a protein called Rho to induce termination and the Second causes termination without the involvement of other factors

Rho-dependent terminators have rather ill-defined RNA elements called **rut sites**. the work of these sites requires the action of the Rho factor

Rho, is a ring-shaped protein with six identical subunits, binds to single-stranded RNA as it exits the polymerase the protein also has an ATPase activity , and once attached to the transcript, Rho uses the energy derived from ATP hydrolysis to induce termination

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Q. Explain the models that shows the functioning of Rho Factors in Bacteria.

i. Rho pushes polymerase forward relative to the DNA and RNA, resulting in termination in a manner analogous to the termination by the protein TRCF(Transcription-repair coupling factor) Rho binds double-stranded DNA upstream of the polymerase and translocates along the DNA until it collides the RNA polymerase the collision pushes polymerase forward, causing dissociation of the ternary complex of RNA polymerase, template DNA and RNA transcript.

ii. Rho pulls RNA out of the polymerase, resulting in termination

iii. Rho induces a conformational change in polymerase, causing the enzyme to terminate the transcription.

Q. Why Rho-independent called intrinsic terminators?

Rho-independent terminators are also called intrinsic terminators because they need no other factors to work

Q. Define Hairpin

Ans. When polymerase trancribes an inverted repeat sequence, the resulting RNA can form a stem-loop structure often called a hairpin by base -paring with itself. formation of the hairpin causes termination by disrupting the elongation complex

Q. what factors required for transcription in eukaryotes?

Ans Additional factors required DNA binding regulatory proteins, mediator complex and chromatin-modifying enzymes.

Q. in eukaryotes transcription What elements found in Pol II core promoter?

Ans The Elements found in pol II promoter include the TFIIB recognation element(BRE), the TATA box, the Initiator (Inr), and the downstream promoter elements known as DPE, DCE and MTE).

Q what is the funtion of GTFS?

Ans The general transcription factors help polymerase bind to the promoter and melt the DNA and they also help polymerase escape from the promoter and embark on the elongation phase

Q. Define Pre-Initiation complex? in eukaryotes transcription.

Ans The complete set of general transcription factors and polymerase bound together at the promoter and poised for initiation, is called the pre-initiation complex.

Q in eukaryotes transcription the TATA elmenet is recongnised by?

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Ans The TATA element recognized by the general transcription factor called TFIID and TFII denotes a transcription factor for pol II

Q. Define TBP?

Ans the component of TFIID that binds to the TATA DNA sequence is called TBP(Tata binding Protein)

Q. write the promoter escape steps in eukaryotes.

Ans in eukaryotes, promoter escape involves two steps. one is ATP hydrolysis and the other is phosphorylation of the polymerase.

Q what is the significance of phosphorylation in promoter Escape.

Ans Addition of these phosphates helps polymerase shed most of the general transcription factors used for initiation and which the enzyme leaves behind as it escapes the promoter. Regulating the phosphorylationn state of the CTD of Pol II controls subsequent steps-elongation and ever procession of the RNA- as well.

Q. Define the funtion of TFIIB .

Ans This protein, a single polypeptide chain, enters the preinitiation complex after TBP. the crystal structure of the ternary complex of TFIIB-TBP-DNA shows specific TFIIB-TBP and TFIIB-DNA contacts. This protein appears to bridge the TATA-Bound TBP and Polymerase.

Q. Write function of TFIIE

ans TFIIE consists of two subunits , binds next and has roles in the recruitment and regulation of TFIIH

Q. Write the function of TFIIH.

Ans TFIIH controls the ATP-dependent transition of the preinitiation complex to the open complex. it is the largest and most complex of the general transcription factor having 10 subunits and a molecular mass comparable to that of the polymerase itself

Q. how does TFIIH mediate promoter melting?

Ans It is now believed that a subunit of TFIIH acts as an ATP-driven translocator of double-stranded DNA this subunit binds to DNA downstream from polymerase and feeds double-stranded DNA, with a right-handed threading into the cleft of the polymerase. This action drives the melting of the DNA becuase the upstream promoter DNA is held in a fixed position by TFIID and the rest of the GTFs.

Q. What is called activators?

Ans Transcriptional regulatory proteins called activators help recruit polymerase to the promoter, stabilizing its binding there

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Q. write the name of RNA elongation factors?

Ans is TFIIS and SPT5 are elongation factors that stimulate elongation

Q. how the first Human ELL protein was identified?

Ans the first human ELL protein was orginally identified as the product of a gene that undergoes translocation in acute myeloid leukemia.

Q. how TFIIS perform function as proofreading ?

Ans TFIIS stimulates an inherent RNAse activity in polymerase allowing an alternative approach to removing misincorporated bases through local limited RNA degradation. This feature is comparable to the hydrolytic editing in the bacterial case stimulated by the Gre factors

Q What is called the Torpedo Model of Termination?

Ans An enzymes that degrades the second RNA as it emerges from the polymerase has been identified and this enzyme may itself trigger termination this is called the torpedo model of termination.

Q. Explain Allosteric Model.?

Ans According to this model termination depends on a conformational change in the elongating polyerase that reduces the processivity of the enzyme leading to sopontaneous termination soon afterward. this conformational change would be linked to polyadenylation and could be triggered by the transfer of the 3` processin enzymes from the CTD tail of polymerase to the RNA

Q. When does the Pol I is required?

Ans. Pol I is required for the expression of only one gene, that encoding the rRNA precursor.

Q. Define Codon?

Ans The coding sequence of a gene is a series of three nucleotides codons that specifies the linear sequence of amino acids in its polypeptide product.

Q. what is difference between exons and introns?

Ans The coding sequences are called **exons** and the intervening sequences are called **introns.**

Q. Defien RNA splicing ?

The primary transcripts of intron-containing genes must have theri introns removed before they can be translated into proteins the process of intron removal is called RNA splicing it converts the pre-mRNA into mature mRNA contain only exons

Q. Why RNA splicing occurs with great precision?

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ans RNA Splicing must occur with great precision to avoid the loss or addition of even a single nucleotide at the sites at which the exons are joined. the triplet-nucleotide codons of mRNA translated in a fixed reading frame that is set by the first codon so lack of precision in splicing will change the reading frames of exons

Q. Define Alternative splicing?

Ans Alternative splicing strategy enables a gene to give rise to more than one polypeptide product these alternative products are called isoforms

Define the structure and function of Spliceosome?

Ans the transestrificatoin reaction are mediated by a huge molecular machine called the spliceosome this complex comrises about 150 proteins and 5 RNAs and is similar in size to a ribosome in performing even a single splicing reaction, the spliceosome hydrolyzes several molecules of ATP

Q. what is called snRNPs

the small nuclear RNAs (snRNAs) each of these RNAs is between 100 to 300 nucleotides long in most eukaryotes and is complexed with several proteins these RNA-Protein complexes are called small nuclear Ribonuclear protiens (snRNPs)

Q write the role of Spliceosome?

Ans There are 3 roles of spliceosome

1) they recognize the 5` splice stie and the branch site
2) they bring those sites together as required and
3) they catalyze the RNA cleavage and joining reactions

Q writes the name of classes of splicing found in cells?

Ans There are total three classes of splicing found in the cells

1) Nuclear pre-mRNA 2) Group II introns 3) Group I introns

Q. What is called Self-splicing?

Ans Self-splicing mean that the intron itself folds into a specific conformation within the precursor RNA and catalyzes the chemistry of its own release. OR

Self-splicing refers to introns that can remove themselves from RNAs in the test tube in the absence of any proteins or other RNA molecules. Group I and II are self-splicing introns

Q. What kind of errors splice-site recognition is prone?

Ans Splice-site recognition is prone to two kinds of errors

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1) first splice sites can be skipped, with components bound at for example a given 5` splice site paring with those at a 3` site beyond the correct one.
2) second, other sites close in sequence but not legitimate splice sites, could be mistakenly recognized because the splice site consensus sequence are rather loose.

therefore, for example components at a given 5` splice site might pair with components bound incorrectly at such a "pseudo" 3` splice site.

Q. What ways the accuracy of splice-site selection can be enhanced?

Ans First, while transcribing a gene the produce the RNA, RNA polymerase II carries with it carious proteins with roles in RNA procession these include proteins involved in splicing. When a 5` splice site is encountered in the newly synthesized RNA, the factors that recognize that site are transferred from the polymerase carboxyl-terminal tail onto the RNA. once in place the 51 splice site components are poised to interact with those other factors that bind to the next 3` splice site to be synthesized. Thus the correct 3` splice site can be recognized before any competing sites further downstream have been transcribed. This co-transcriptional loading process greatly diminishes the likelihood of exon skipping. SR proteins bound to these sites interact with the components of the splicing machinery, recruiting them to the nearby splice sites. SR proteins are essential for splicing and they not only ensure the accuracy and efficiency of constitutive splicing but also regulate alternative splicing.

Q. Define Minor spliceosome? And its function

Ans The identical U5 component is found in both the major and the alternative splicing is so called minor spliceosome. the minor spliceosome recognizes rarely occurring introns having consensus sequences distinct from the sequences of most pre-mRNA introns

Q in what ways the alternative splicing can occurs?

Ans. Alternating splicing can occur in a number of ways. thus in addition to alternative exons, exons can be extended. in other cases, exons can be skipped or introns can be retained in the mature message. Some alternative splicing results from transcription of a gene from alternative promoters, allowing one transcript to include a 5` exon not present in the other

Q. What is called Cassette Exons?

Ans. in genome-wide studies, the most commonly seen forms of alternative splicing are cases in which complete exons are included or excluded from the mature message. such exons are often called cassette exons.

Q. Write the names of mechanisms that ensure that the selection of alternative exons is mutually exclusive?

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Ans there are 3 mechanisms 1) Steric Hindrance 2) combination of major and Minor splice sites 3) nonsense-mediated Decay

Q. Define enhancers or silencers?

Ans Proteins that regulate splicing bind to specific sites called exonic or intronic splicing enhancers ESE or ISE or Silencers ESS or ISS

Q. what is function of hnRNPI protein?

Ans. In some cases this protein blocks the binding of the basic splicing machinery by binding directly to the Py tract (hnRNPI is also called the polypyrimidine tract-binding protein). in other cases it excludes a given exon from the mature mRNA by binding to sequences that flank that exon

Q. Explain how alternative splicing is used for switching expression of the genes on or off?

Ans switching expression of the genes on and off this is achieved in two ways. Most straight farwardly, an exon contains a stop codon and when incorporated into mRNA this prematurely terminates translation generating a truncated polypeptide. the second way alternative splicing can be used as an on/off switch is by regulating the use of an intron which when retained in the mRNA ensure that species is not transported out of the nucleus and so is never translated

Q. Define deamination of adenosine?

Asn RNA editing take place through demination of adenosine which converts adenosine to inosine which as an oxygen in place of adenine's amino group because inosine forms base pairs with cytidine in the same way as guanosine the deamination of adenosine changes the meaning of a codon forexample an ACG threonine codon becomes ICG codon, which would be read by the riobsome as GCG (alanine)

Q. How are RNA selection and transport achieved?

Ans from the moment an RNA molecule starts to be transcribed; it becomes associated with proteins of various sorts. initially proteins involved in capping, then splicing factors and finally the proteins that mediate polyadenylation some of these proteins are replaced at various steps along the processing path but others are not and moreover additional proteins join as a result, a typical mature mRNA carries a collection of proteins that identifies it as being mRNA destined for transport.

Q. What is the role of hnRNPs in mRNA transport?

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Ans The mRNA do aslo contain some hnRNPs, but fewer than are typically bound to introns and in a different context as well this emphasizes the fact that it is the set of proteins, not any individual kind of protein that marks RNAs for either export or retention in the nucleus

Q What is called translation?

Ans The genetic information contained with the order of nucleotides in mRNA as interpreted to generate the linear sequences of aminoacids in proteins this process is known as translation .

Q What are the primary components involved in translating the language of mRNAs into protein language?

Ans The machinery responsible for translating the language of mRNAs into the language of proteins is composed of four primary components mRNAs, tRNAs , Aminoaceyl-tRNA synthetases and the Ribosomes.

Q. what is called ORF?

Ans The protein-coding regions of each mRNA is composed of a contiguous, non-overlaping string of codons called an open reading frame known as ORF

Q. What are the functions of start codon?

Ans The start codon has two important functions 1) first it specifies the first amino acid to be incorporated into the growing polypeptide chain 2) second it defines the reading frame for all subsequent codons

Q what are the stop codons?

Ans there are three stop codons i.e UAG. UGA and UAA

Q what is difference between polycistronic RNAs and MOnocistronic RNAs?

Ans. prokaryotic mRNAs frequently contain two or more ORFs. mRNAs containing multiple ORFs are known as polycistronic RNAs and those encoding a single ORF are known as monocistronic RNAs.

Q. Define the function of Polycistronic?

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Ans The polycistronic mRNAs found in bacteria often encode proteins that perform related functions such as different steps in the biosynthesis of an aminoacid or nucleotide.

Q what is Called RBS?

Ans To facilitate binding by a ribosome many prokaryotic ORFs contain a short sequence upsrteam of the start codon called the Ribosome-binding site RBS. this element is also referred to as a shine olagarno sequence.

Q what is translational coupling?

Ans a ribosome that has just completed translating the upstream ORF is positioned to begin translating from the start codon for the downstream ORF this phenomenon of linked translation between overlapping ORFs is known as translational coupling

Q. What is called 5` cap?

the 5` cap is a methylated guanine nucleotide that is joined to the 5` end of the mRNA via an unusual 5` to 5` linkage created in three steps the guanine nucleotide of the 5` cap is connected to the 5` end of the mRNA through three phosphate groups.

Q define the process called scanning?

Ans the resulting 5` cap is required to recruit the ribosome to the mRNA. once bound to the mRNA the ribosome moves in a 5` to 3` direction until it encounters a 5` AUG-3` start codon, a process called scanning

Q. what is called heart of protein synthesis?

Ans The heart of protein synthesis is the translation of nucleotide sequence information into amino acids

Q. writes the name of channels of Secondary structure of tRNAs

1) Amino acid attachment site 2) TC loop 3) Anticodon loop 4) DHU loop 5) variable loop

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Question from lecture No.153-178

Q. When a peptide bond formed in ribozymes?

Ans....

When correctly charged tRNA has been placed in the A site and has rotated into the peptidyl transferase center, peptide bond formation takes place

Q. Which interactions are likely stabilize the aminoacyl-tRNA after accommodation?

Ans.....

base-pairing between the 23 S rRNA and the CCA ends of the tRNAs in the A and the P sites help to position the alpha-amino group of the aminoacyl-tRNA to attack the carbonyl group of the growing polypeptide attached to the peptidyl-tRNA These interactions are also likely to stabilize the aminoacyl-tRNA after accommodation

Q.Define and abbreviate EF-G?

Ans.....

Elongation facor

Definition: The completion of translocation requires the action of a second elongation factor called EF-G

Q.Which interactions are use to unlock ribosomes?

Ans.....

Interactions between EF-G–GDP and the ribosome are use to unlock the ribosomes.

(Justification: there are ***gates*** that separate the A-, P-, and E-sites and EF-G–GDP is said to unlock the ribosome by opening these gates)

Q.What happened when ribosomes unlocked?

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Ans.....

Due to ribosomes unlocked formerly A-site tRNA free to move into in the P-site, allowing EF-GDP to bind the A-site.

Q.What are RFs ? And Write their role?

Ans.....

stop codons are recognized by proteins called release factors ***RFs*** that activate the hydrolysis of the polypeptide from the ppeptidyl-tRN

Q.What are the classes of release factors? Write their role in prokaryotes?

Ans....

Prokaryotes have two class I release factors called RF1 and RF2. RF1 recognizes the stop codon UAG and RF2 recognizes the stop codon UGA. The third stop codon, UAA, is recognized by both RF1 and RF2

1) **What formation accure due to the binding of GTP to RF3?**

Ans....

The binding of GTP to RF3 leads to the formation of a high-affinity interaction with the ribosome that favors the rotated hybrid state.

2) **What is trigger factor?**

Ans.....

In bacteria, co-translational protein folding is assisted by a 48 kDa chaperone called the trigger factor which binds at the bacterial ribosome's exit pore

3) **What is result of accumulated toxic proteins?**

Ans.....

accumulation of toxic proteins result from misfolding may lead to several neurological disorders including Alzheimer's, Huntington's and Parkinson's diseases.

4) **What product produced by free ribosomes and membrane-bound ribosomes?**

Ans.....

The free ribosomes synthesize cytoplasmic and

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mitochondrial proteins whereas, the membrane-bound ribosomes synthesize integral membrane proteins, lysosomal proteins and secretary proteins.

5) **What hypothesis proposed by Blobel and Sabitini for proteins synthesis?**

Ans.....

.1) Free and membrane-bound ribosomes are identical.

• 2) Protein synthesis always begins on free ribosomes.

• 3) Nascent secretary, trans-membrane, or lysosomal proteins have sequences of 20-30 amino acids at their amino terminus that act as signals to bind the ribosomes to the endoplasmic reticulum.

6) **Which components are required for SRP cycle?**

Ans.....

SRP having 6 protein

SRP receptors contains two GTP-binding polypeptide Subunits

Translocons

7) **What is translocons?**

Ans.....

The protein conducting channel, which is made of three polypeptides (Sec61, Sec61β and Sec61) acts as a passive conduit for polypeptides

8) **What is role of bacterial regulatory RNAs in eukaryotes?**

Ans.....

bacterial regulatory RNAs (called sRNAs) acts in trans to control translation of target genes, rather as microRNAs do in eukaryotes.

9) **What are the components of riboswitch ?**

Ans.....

The aptamer and, the expression platform are two components of ribiswitch.

10) **TPP and FMN stands for?**

Ans.....

thiamine pyrophosphate (TPP)

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flavin mononucleotide (FMN)

11) What are the sources of ribiswitch?

Ans.....

Riboswitches are found in archaea, fungi, and plants.

12) CRISPRs stands for?

Ans.....

Clustered Regularly Interspaced Short Palindromic Repeats

13) Write names of three regulatory RNAs ?

Ans.....

siRNA small interfaring RNA

miRNA. Micro RNA

piRNA. Piwi interaction RNA

14) What is Dicer write its function?

Ans.....

Dicer, an RNase III–like enzyme that recognizes and digests longer dsRNAs or the stem-loop structures formed by miRNA precursors.

15) RISC stands for?

Ans.....

RNA-induced silencing complex (RISC).

16) How many types of cis-acting elements are there write names?

Ans.....

There are four types of cis-acting elements Promoter, Enhancer, Silencer, Response element.

17) What types of promotors are based on relative location?.

Ans.....

- Class II Promoters
- Class I Promoters
- Class III Promoters

18) DCE stands for?

Ans.....

Downstream core elements

19) What are enhancer and silencers?

Ans....

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Enhancers are elements that stimulate transcription while

Silencers, by contrast, depress transcription.

20) What is ORF? Give abbreviation?

Ans.....

Open reading frame;

The segment of mRNA that codes for a polypeptide chain

is called an open reading frame (ORF)

21) What do you know about cistron and monocistronic RNAs?

Ans.....

A DNA segment corresponding to an ORF plus the

translational start and stop signals for protein synthesis is

called a cistron and the mRNA encoding a single

polypeptide is called monocistronic mRNA

22) Differentiate between 5' UTR and 3'UTR also write their abbreviation?

Ans.....

The untranslated RNA sequence before the coding region

is called the 5' – utranslated region (5'- UTR) or 5' – leader

while the untranslated sequence after the coding region is

called 3'– utranslated region (3'- UTR)

23) Diffrence between negative and positive regulation?

Ans.....

In negative regulation, a repressor turns off the transcription of one or more gene.

In positive regulation, an activator turns on the transcription of one or more gene.

24) Define Inducible enzyme?

Ans.....

Enzymes such as β-galactosidase, permease and

tranacetylase are said to be inducible enzymes

25) IPTG stands for ?

isopropylthio-galactoside

26) What are allolactose?

Ans.....

isolated Lac repressor does not bind lactose

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but does bind a lactose isomer called allolactose.

27) Lac repressor represented by which scientists tell name?

Ans....

Walter Gilbert and Benno Muller-Hill successfully isolated the Lac Repressor from E. coli extracts in 1966

28) What is function of β-galactosidase in lactose metabolism?

Ans.....

The function of the β-galactosidase in lactose metabolism is to hydrolyze lactose to form glucose and galactose

29) Define catabolite repression?

Ans.....

Transcription-level inhibition of the lactose enzymes and a variety of other inducible enzymes by glucose (or other readily used carbon source) is called catabolite repression.

Question from lecture No.179-202

1) What affect seems due to absense of cAMP-CRP complex?

Ans...In the absence of cAMP-CRP complex, the lac promoter is quite weak because its – 10 box differs significantly from the consensus sequence.

In the absence of cAMP, CRP□DNA interactions are non specific and weak

2) AS stands for?

Ans....activator site

3) When a cAMP-CRP sequence permanently switched on?

Ans.... Binding of cAMP□CRP to the consensus sequence is so tight that an operon containing this sequence would be permanently switched on

4) How many genes a structural operon contains? Mention their name?

Ans....sensitive operon, contains four structural genes galK, galT, galE and galM which specifies the enzymes galactokinase, galactose transferase, galactose epimerase & mutarotase

5) Explain gel operon's operators?

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Ans..... The gal operon has two operators, galOE (OE) and galOl (Ol) One operators, galOE , is upstream from the promoter region and the other galOl is in galE.

6) **What do you know about HU?**

Ans.....Sanker Adhya and coworkers solved the problem by showing that a histone-like protein called HU binds to the gal promoter region only when the Gal repressors are bound at both galOE or galOI causing the DNA to loop.

7) **Define class II cAMP-dependant promoters?**

Ans..... Promoters like P1 in gal operon that have AS at position –41.5 are called class II cAMP-dependant promoters

8) **Why does the gal operon has two promoters?**

@Ans....

Because glactose have two roles in cellular metabolism and When galactose is not available in the growth medium, cells require the epimerase specified by the gal operon to convert glucose to galactose, which is then used to make lipopolysachharides.

9) **What is role of P^1 and P^2 promoters?**

Ans ... P2) is needed for background constitutive synthesis and a cAMP□CRP – dependent promoter. P1) is needed to regulate high-level synthesis.

10) **Briefly describe the structure of tryptophan operon?**

Ans..... The trp operon consists of a promoter, an operator, a leader (trpL), an attenuator, and five structural genes designated as trpE, trpD, trpC, trpB and trpA.

11) **What is the function of TrpR-tryptophane?**

Ans..... The TrpR□tryptophane complex functions as a coarse on-off switch that turns the tryptophane operon off when tryptophane levels are high.

12) **What was the result of Charles Yanofsky's experiments?**

Ans ... an experiment performed by Charles Yanofsky in 1972, which showed that E. coli

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mutants that lack a functional TrpR protein increase trp operon transcription after being starved for tryptophan

13) trpL stands for?

Ans..... tryptophan leader

14) What is attenuator?

Ans..... Constitutive mutants exhibit a sixfold increase in tryptophane enzyme synthesis when bases 123 – 150 within the leader are deleted This 28 base sequence is called the attenuator.

15) What is the function of attenuator?

Ans.....The attenuator can fold into a stem-and-loop structure with the potential to function as a rho-independent transcription terminator.

16) What happened when concentration of tryptophan is very low in RNA polymerase enzyme?

@ns..... In the second case when the concentration of tryptophan is very low, the bulky ribosome will pause at the tryptophane codon (UGG) on segment 1, preventing segment 1 from pairing with segment 2 to form the 1□2 hairpin.

17) What is repressor and activator?

Ans..... Extracellular signals are communicated to genes by regulatory proteins, which come in two types: positive regulators, or activators, and negative regulators, or repressors

18) Why then is not all regulation focused on the step of transcription initiation?

Ans..... Regulating later steps can have two advantages. First, it allows for more inputs: if a gene is regulated at more than one step, more signals can modulate its expression, or the same signals can do so even more effectively. Second, regulation at steps later than transcription initiation can reduce the response time.

19) Define an operator?

Ans..... The site on DNA where a repressor binds is called an Operator

20) How can initiation of transcription stimulates in eukaryotes?

Ans....an activator interacts with polymerase already bound to the

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promoter and, by inducing a conformational change in the enzyme or the promoter, stimulates transcription initiation.

21) Define catabolite repression?

Ans..... Transcription-level inhibition of the lactose enzymes and a variety of other inducible enzymes by glucose (or other readily used carbon source) is called catabolite repression.

22) CAP and CRP stands for?

Ans.... Catabolite activator protein
cAMP recepter proteins

23) PTS stands for?

Ans..... phosphoenol pyruvate-dependant phosphotransferase system (PTS).

24) What is inducer exclusion?

Ans.... mechanism by which glucose influences lac operon transcription is called inducer exclusion, which also involves the glucose transport system.

25) What is CTD also write its function?

Ans..... carboxy-terminal domain (CTD) largest subunit of RNA polymerase, freeing the enzyme from the stall and allowing transcription to proceed through the gene.

26) How can a HIV Viruse (cause AIDS) transcribes its genes,?

Ans.... human immunodeficiency virus (HIV), which causes AIDS, transcribes its genes from a promoter controlled by P-TEFb.

27) Define silencing?

Ans.... Histone (and DNA) modifications also form the basis of a type of repression called silencing

28) What is STAT and MAPK

Ans..... STAT (signal transducer and activator of transcription) pathway is a relatively simple example of signal transduction pathway.mitogen-activated protein kinase (MAPK) pathway that controls activators such as Jun, one of the activators that works at the interferon-β enhancer.

29) What are plasmids?

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Ans.... Plasmids are autonomous self-replicating molecules of DNA (or very rarely RNA)

30) Define cloning and cell clone?

Ans.... process of producing a large quantity of identical copies of any chosen DNA is referred to as cloning, and the cells derived from a host cell containing a recombinant plasmid is referred to as a cell clone.

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